

UNIT-II

Physical Layer

The physical layer has complex tasks to perform. One major task is to provide services for the data link layer. The data in the data link layer consists of Os and Is organized into frames that are ready to be sent across the transmission medium. This stream of Os and Is must first be converted into another entity: signals. One of the services provided by the physical layer is to create a signal that represents this stream of bits.

The physical layer must also take care of the physical network, the transmission medium. The transmission medium is a passive entity; it has no internal program or logic for control like other layers. The transmission medium must be controlled by the physical layer. The physical layer decides on the directions of data flow. The physical layer decides on the number of logical channels for transporting data coming from different sources.

Functions Performed by Physical Layer

The following are some important and basic functions that are performed by the Physical Layer of the OSI Model –

1. The physical layer maintains the data rate (how many bits a sender can send per second).
2. It performs the Synchronization of bits.
3. It helps in Transmission Medium decisions (direction of data transfer).
4. It helps in Physical Topology (Mesh, Star, Bus, Ring) decisions (Topology through which we can connect the devices with each other).
5. It helps in providing Physical Medium and Interface decisions.
6. It provides two types of configuration Point Point configuration and Multi-Point configuration.
7. It provides an interface between devices (like PCs or computers) and transmission medium.
8. It has a protocol data unit in bits.
9. Hubs, Ethernet, etc. device is used in this layer.
10. This layer comes under the category of Hardware Layers (since the hardware layer is responsible for all the physical connection establishment and processing too).
11. It provides an important aspect called Modulation, which is the process of converting the data into radio waves by adding the information to an electrical or optical nerve signal.
12. It also provides a Switching mechanism wherein data packets can be forwarded from one port (sender port) to the leading destination port.

DATA RATE LIMITS:

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second. According to the formula, we might think that, given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels. Although the idea is theoretically correct, practically there is a limit. When we increase the number of signal levels, we impose a burden on the receiver. If the number of levels in a signal is just 2, the receiver can easily distinguish between a 0 and a 1. If the level of a signal is 64, the receiver must be very sophisticated to distinguish between 64 different levels. In other words, increasing the levels of a signal reduces the reliability of the system.

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Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel: $\text{Capacity} = \text{bandwidth} \times \log_2 (1 + \text{SNR})$. In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second. Note that in the Shannon formula there is no indication of the signal level, which means that no matter how many levels we have, we cannot achieve a data rate

higher than the capacity of the channel. In other words, the formula defines a characteristic of the channel, not the method of transmission.

Transmission media:

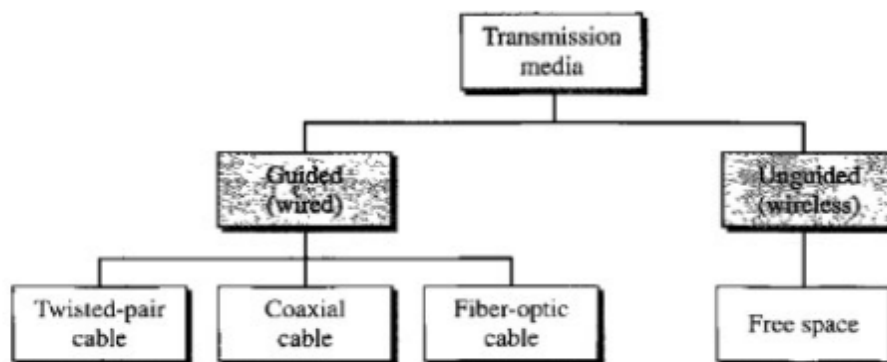
Transmission media are actually located below the physical layer and are directly controlled by the physical layer. that transmission media belongs to layer zero. Figure shows the position of transmission media in relation to the physical layer.

Transmission Medium: A transmission medium can be broadly defined as anything that can carry information from a source to a destination.

For example, the transmission medium for two people having a dinner conversation is the air. The air can also be used to convey the message in a smoke signal or semaphore.

For a written message, the transmission medium might be a mail carrier, a truck, or an airplane.

In data communications the definition of the information and the transmission medium is more specific. The transmission medium is usually free space, metallic cable, or fiber-optic cable. The information is usually a signal that is the result of a conversion of data from another form.



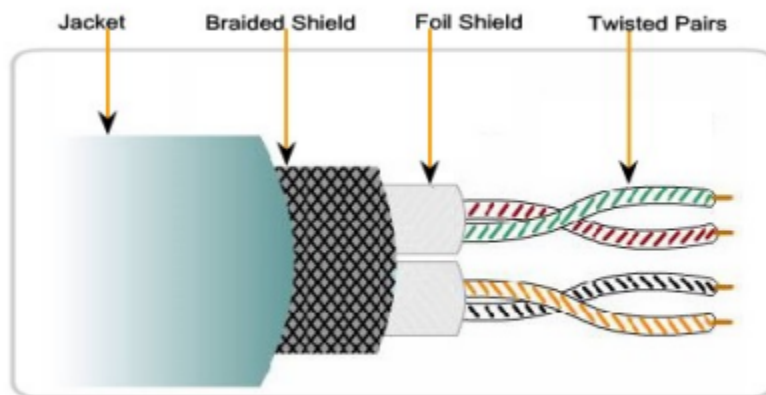
Guided Media:

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the medium. Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Optical fiber is a cable that accepts and transports signals in the form of light.

Shielded twisted-pair (STP):

Another type of cabling used in networking is shielded twisted-pair (STP). As shown in the figure, STP uses two pairs of wires that are wrapped in an overall metallic braid or foil. STP

cable shields the entire bundle of wires within the cable as well as the individual wire pairs. STP provides better noise protection than UTP cabling, however at a significantly higher price. For many years, STP was the cabling structure specified for use in Token Ring network installations. With the use of Token Ring declining, the demand for shielded twisted-pair cabling has also waned. The new 10 GB standard for Ethernet has a provision for the use of STP cabling. This may provide a renewed interest in shielded twisted-pair cabling.

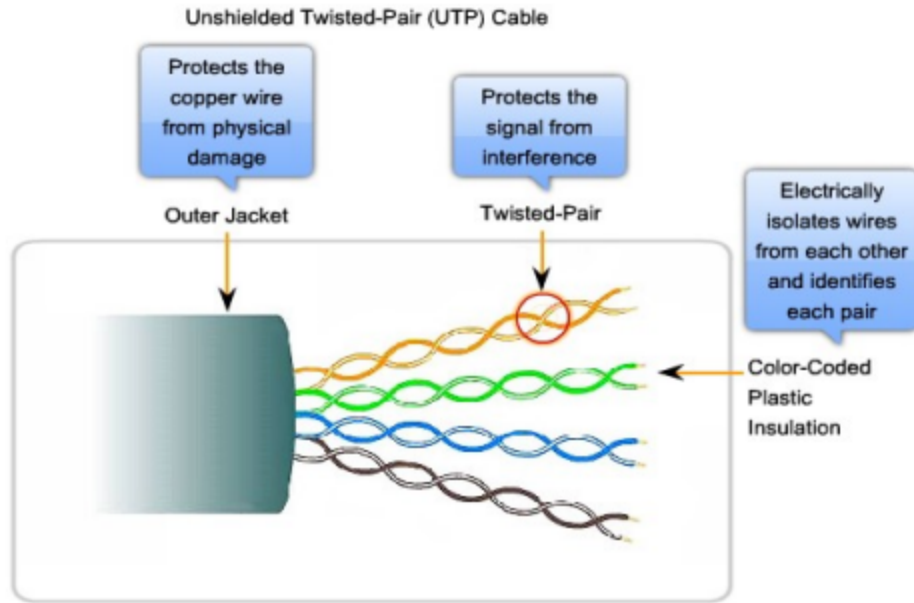


Shielded Twisted-Pair Cable(STP)

Unshielded twisted-pair (UTP):

Unshielded twisted-pair (UTP) cabling, as it is used in Ethernet LANs, consists of four pairs of color-coded wires that have been twisted together and then encased in a flexible plastic sheath. As seen in the figure, the color codes identify the individual pairs and wires in the pairs and aid in cable termination.

The twisting has the effect of canceling unwanted signals. When two wires in an electrical circuit are placed close together, external electromagnetic fields create the same interference in each wire. The pairs are twisted to keep the wires in as close proximity as is physically possible. When this common interference is present on the wires in a twisted pair, the receiver processes it in equal yet opposite ways. As a result, the signals caused by electromagnetic interference from external sources are effectively cancelled. This cancellation effect also helps avoid interference from internal sources called crosstalk. Crosstalk is the interference caused by the magnetic field around the adjacent pairs of wires in the cable. When electrical current flows through a wire, it creates a circular magnetic field around the wire. With the current flowing in opposite directions in the two wires in a pair, the magnetic fields - as equal but opposite forces - have a cancellation effect on each other. Additionally, the different pairs of wires that are twisted in the cable use a different number of twists per meter to help protect the cable from crosstalk between pairs.



Applications : Twisted-pair cables are used in telephone lines to provide voice and data channels. The local loop-the line that connects subscribers to the central telephone office---commonly consists of unshielded twisted-pair cables.

The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.

UTP Cabling Standards:

The UTP cabling commonly found in workplaces, schools, and homes conforms to the standards established jointly by the Telecommunications Industry Association (TIA) and the Electronics Industries Alliance (EIA). TIA/EIA-568A stipulates the commercial cabling standards for LAN installations and is the standard most commonly used in LAN cabling environments.

Some of the elements defined are:

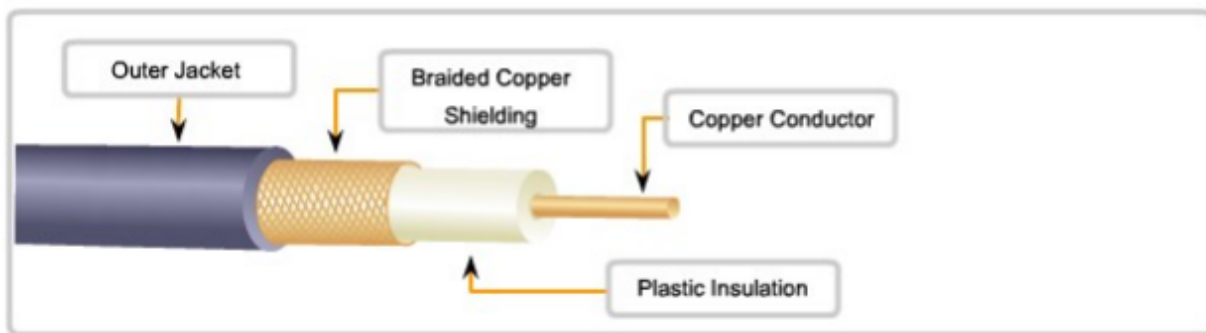
- Cable types
- Cable lengths
- Connectors
- Cable termination
- Methods of testing cable

The electrical characteristics of copper cabling are defined by the Institute of Electrical and Electronics Engineers (IEEE). IEEE rates UTP cabling according to its performance. Cables are placed into categories according to their ability to carry higher bandwidth rates. For example, Category 5 (Cat5) cable is used commonly in 100BASE-TX FastEthernet installations. Other

categories include Enhanced Category 5 (Cat5e) cable and Category 6 (Cat6). Cables in higher categories are designed and constructed to support higher data rates. As new gigabit speed Ethernet technologies are being developed and adopted, Cat5e is now the minimally acceptable cable type, with Cat6 being the recommended type for new building installations.

Co-axial Cable:

Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted-pair cable, in part because the two media are constructed quite differently. Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.



All the elements of the coaxial cable encircle the center conductor. Because they all share the same axis, this construction is called coaxial, or coax for short.

Uses of Coaxial Cable :

The coaxial cable design has been adapted for different purposes. Coax is an important type of cable that is used in wireless and cable access technologies. Coax cables are used to attach antennas to wireless devices. The coaxial cable carries radio frequency (RF) energy between the antennas and the radio equipment.

Coax is also the most widely used media for transporting high radio frequency signals over wire, especially cable television signals. Traditional cable television, exclusively transmitting in one direction, was composed completely of coax cable. Cable service providers are currently converting their one-way systems to two-way systems to provide Internet connectivity to their customers. To provide these services, portions of the coaxial cable and supporting amplification elements are replaced with multi-fiber-optic cable. However, the final connection to the customer's location and the wiring inside the customer's premises is still coax cable. This combined use of fiber and coax is referred to as hybrid fiber coax (HFC).

In the past, coaxial cable was used in Ethernet installations. Today UTP offers lower costs and higher bandwidth than coaxial and has replaced it as the standard for all Ethernet installations.

Coaxial Cable Connectors To connect coaxial cable to devices, we need coaxial connectors. The most common type of connector used today is the Bayone-Neill-Concelman (BNC), connector.

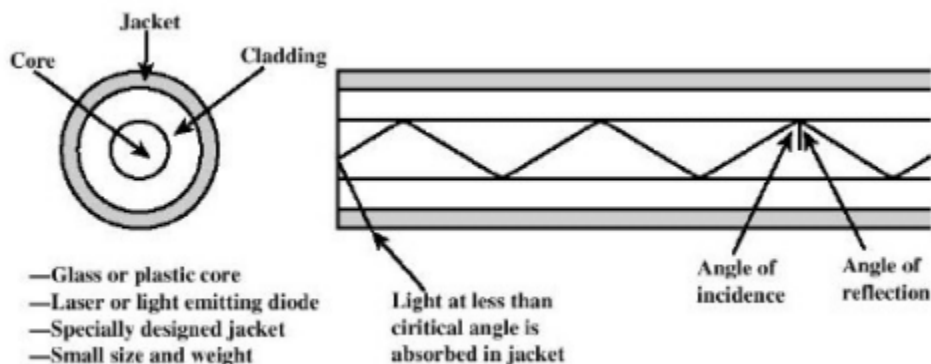
Three types of connectors: the BNC connector, the BNC T connector, and the BNC terminator. The BNC connector is used to connect the end of the cable to a device, such as a TV set. The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device. The BNC terminator is used at the end of the cable to prevent the reflection of the signal.

Fiber-optics:

Fiber-optic cabling uses either glass or plastic fibers to guide light impulses from source to destination. The bits are encoded on the fiber as light impulses. Optical fiber cabling is capable of very large raw data bandwidth rates. Most current transmission standards have yet to approach the potential bandwidth of this media.

Principle of Fiber-optics:

It is based on the principle of Total internal Reflection.



Advantages and Disadvantages of Optical Fiber:

Advantages

Fiber-optic cable has several advantages over metallic cable (twisted pair or coaxial).

Higher bandwidth: Fiber-optic cable can support dramatically higher bandwidths (and hence data rates) than either twisted-pair or coaxial cable. Currently, data rates and bandwidth utilization over fiber-optic cable are limited not by the medium but by the signal generation and reception technology available.

Less signal attenuation:

Fiber-optic transmission distance is significantly greater than that of other guided media. A signal can run for 50 km without requiring regeneration. We need repeaters every 5 km for coaxial or twisted-pair cable.

Immunity to electromagnetic interference:

Electromagnetic noise cannot affect fiber-optic cables. o Resistance to corrosive materials. Glass is more resistant to corrosive materials than copper

Resistance to corrosive materials. Glass is more resistant to corrosive materials than copper.

Light weight. Fiber-optic cables are much lighter than copper cables

Greater immunity to tapping. Fiber-optic cables are more immune to tapping than copper cables. Copper cables create antenna effects that can easily be tapped.

Disadvantages

There are some disadvantages in the use of optical fiber.

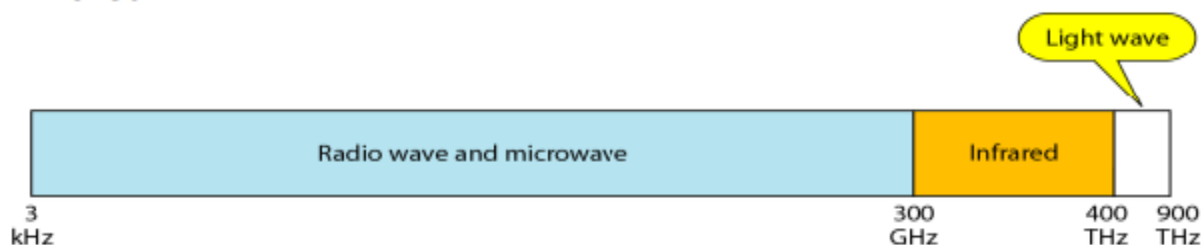
Installation and maintenance. Fiber-optic cable is a relatively new technology. Its installation and maintenance require expertise that is not yet available everywhere.

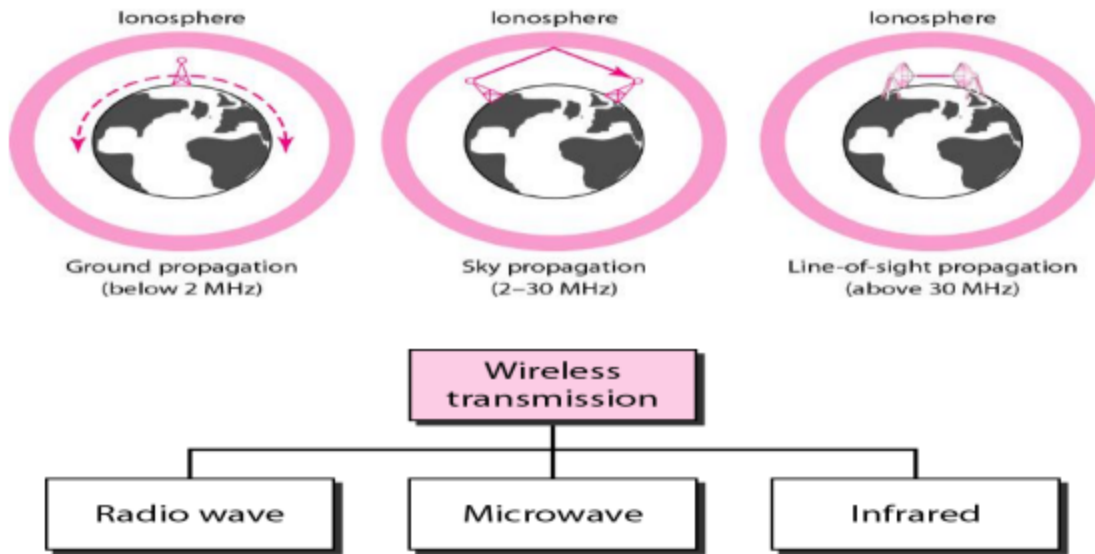
Unidirectional light propagation. Propagation of light is unidirectional. If we need bidirectional communication, two fibers are needed.

Cost. The cable and the interfaces are relatively more expensive than those of other guided media. If the demand for bandwidth is not high, often the use of optical fiber cannot be justified

Unguided media: (Wireless)

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication. Radio Waves Microwaves Infrared





Radio Waves : Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves. Radio waves are omni directional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna. The omni directional property has a disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band. **Omni directional Antenna :**

Radio waves use omnidirectional antennas that send out signals in all directions. Based on the wavelength, strength, and the purpose of transmission, we can have several types of antennas. Figure shows an omnidirectional antenna



Applications :

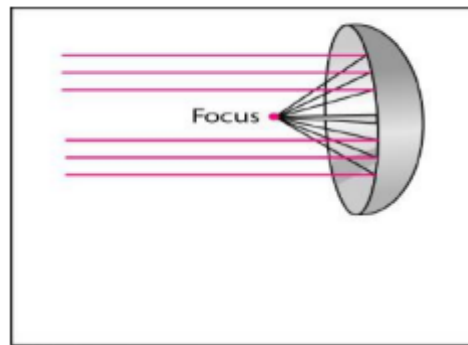
The Omni directional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

Microwaves

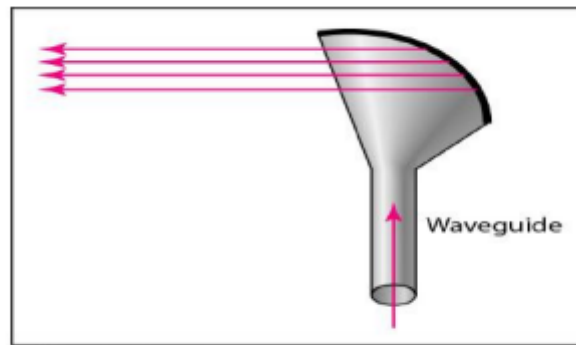
Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. The sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas

Unidirectional Antenna :

Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish and the horn. Applications: Microwaves are used for unicast communication such as cellular telephone



a. Dish antenna



b. Horn antenna

Applications:

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

Infrared:

Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a shortrange communication system in one room cannot be affected by another system in the next room. When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. Infrared signals are useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Applications:

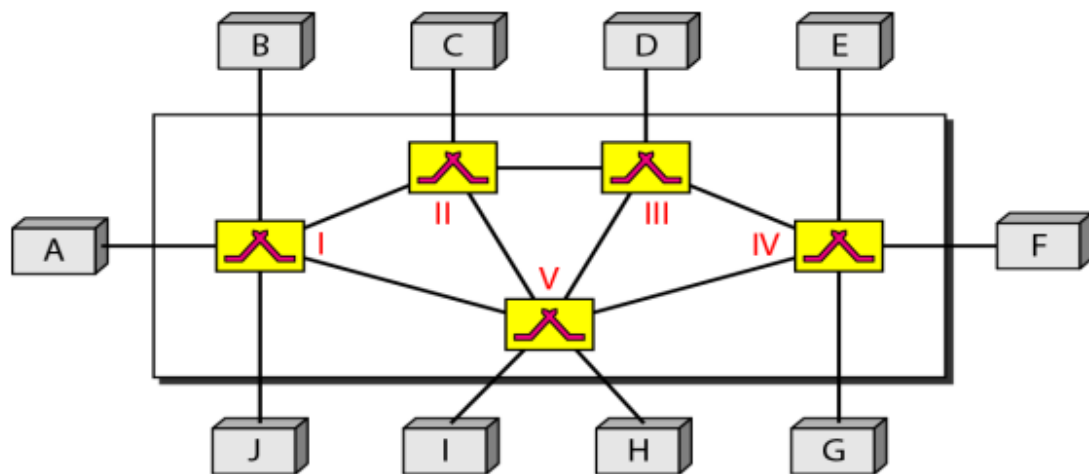
Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Switching systems:

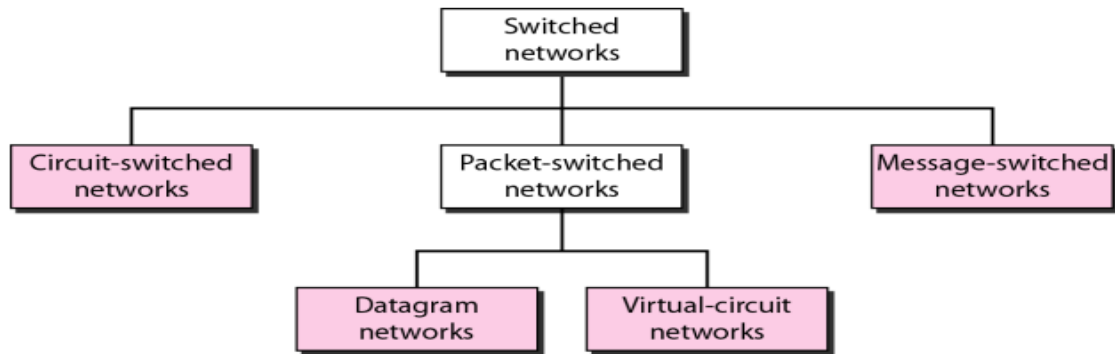
A network is a set of connected devices. Whenever we have multiple devices, we have the problem of how to connect them to make one-to-one communication possible. One solution is to make a point-to-point connection between each pair of devices (a mesh topology) or between a central device and every other device (a star topology). These methods, however, are impractical and wasteful when applied to very large networks.

The number and length of the links require too much infrastructure to be cost-efficient, and the majority of those links would be idle most of the time.

A better solution is switching. A switched network consists of a series of interlinked nodes, called switches. Switches are devices capable of creating temporary connections between two or more devices linked to the switch. In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing. Figure shows a switched network.

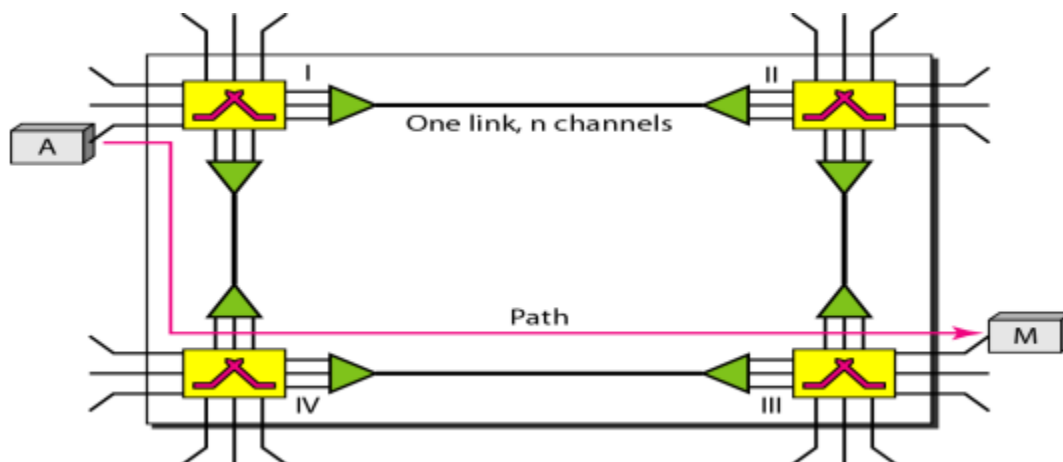


We can then divide today's networks into three broad categories: circuit-switched networks, packet-switched networks, and message-switched. Packet-switched networks can further be divided into two subcategories-virtual-circuit networks and datagram networks



CIRCUIT-SWITCHED NETWORKS:

A circuit-switched network consists of a set of switches connected by physical links. A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link. Each link is normally divided into n channels by using FDM or TDM. In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.



Three Phases The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown.

Setup Phase :

Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established. Connection setup means creating dedicated channels between the switches. For example, in Figure, when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I. Switch I finds a channel between itself and switch IV that can be dedicated for this purpose.

Switch I then sends the request to switch IV, which finds a dedicated channel between itself and switch III. Switch III informs system M of system A's intention at this time.

In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A. Only after system A receives this acknowledgment is the connection established.

Data Transfer Phase :

After the establishment of the dedicated circuit (channels), the two parties can transfer data.

Teardown Phase:

When one of the parties needs to disconnect, a signal is sent to each switch to release the resources

Efficiency :

It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections.

Delay :

Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection. The total delay is due to the time needed to create the connection, transfer data, and disconnect the circuit.

Switching at the physical layer in the traditional telephone network uses the circuit-switching approach.

DATAGRAM NETWORKS:

In a packet-switched network, there is no resource reservation; resources are allocated on demand. The allocation is done on a first come, first-served basis. When a switch receives a packet, no matter what is the source or destination, the packet must wait if there are other packets being processed. This lack of reservation may create delay. For example, if we do not have a reservation at a restaurant, we might have to wait.

In a datagram network, each packet is treated independently of all others. Packets in this approach are referred to as datagrams. Datagram switching is normally done at the network layer

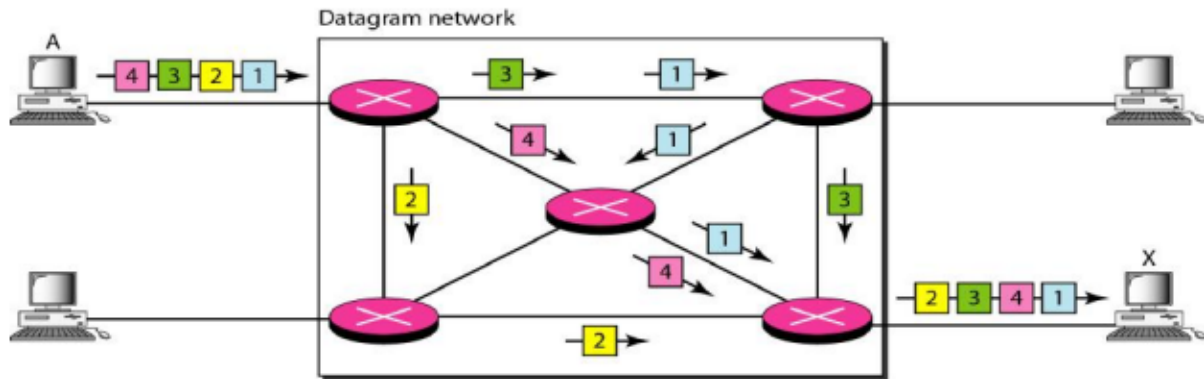


Figure shows how the datagram approach is used to deliver four packets from station A to station X. The switches in a datagram network are traditionally referred to as routers.

The datagram networks are sometimes referred to as connectionless networks. The term connectionless here means that the switch (packet switch) does not keep information about the connection state. There are no setup or teardown phases. Each packet is treated the same by a switch regardless of its source or destination.

A switch in a datagram network uses a routing table that is based on the destination address. The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Efficiency:

The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred.

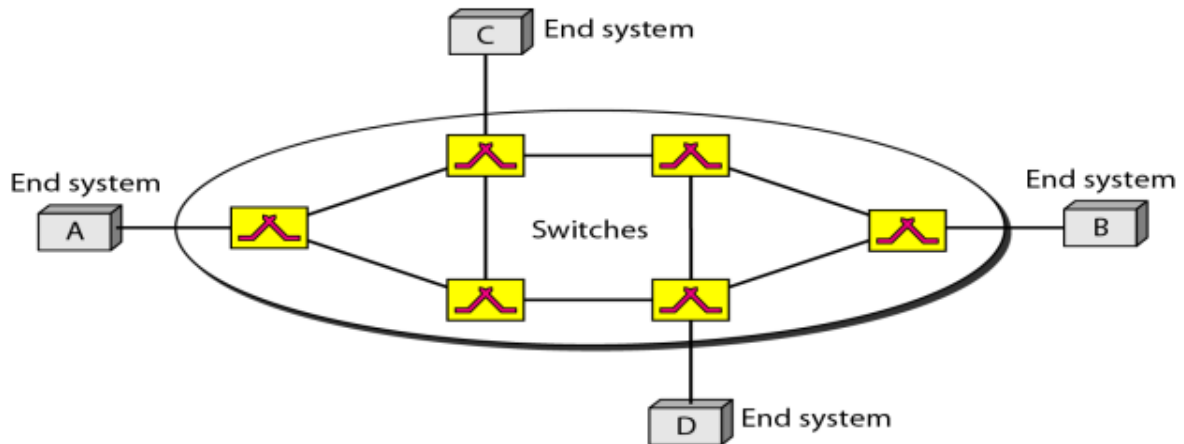
Delay:

There may be greater delay in a datagram network than in a virtual-circuit network. Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded. In addition, since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.

Switching in the Internet is done by using the datagram approach to packet switching at the network layer

VIRTUAL-CIRCUIT NETWORKS:

A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.



1. As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.
2. Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
3. As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has local jurisdiction (it defines what should be the next switch and the channel on which the packet is being carried), not end-to-end jurisdiction.
4. As in a circuit-switched network, all packets follow the same path established during the connection.
5. A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer.

Addressing :

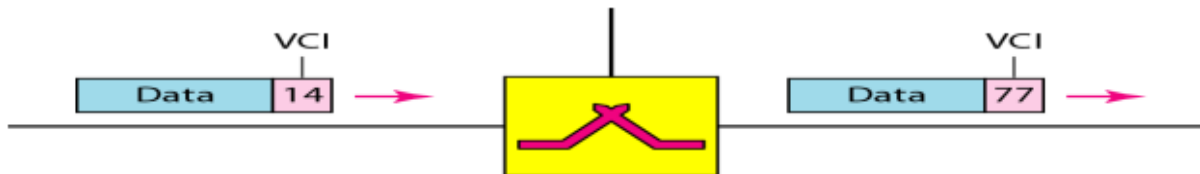
In a virtual-circuit network, two types of addressing are involved: global and local (virtual-circuit identifier).

Global Addressing :

A source or a destination needs to have a global address—an address that can be unique in the scope of the network

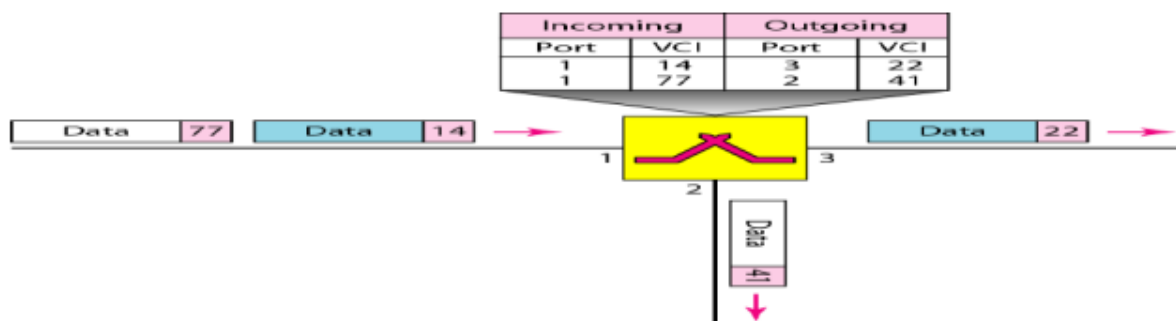
Virtual-Circuit Identifier :

The identifier that is actually used for data transfer is called the virtual-circuit identifier (VCI). A VCI, unlike a global address, is a small number that has only switch scope; it is used by a frame between two switches. When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI. Figure shows how the VCI in a data frame changes from one switch to another. Note that a VCI does not need to be a large number since each switch can use its own unique set of VCIs.

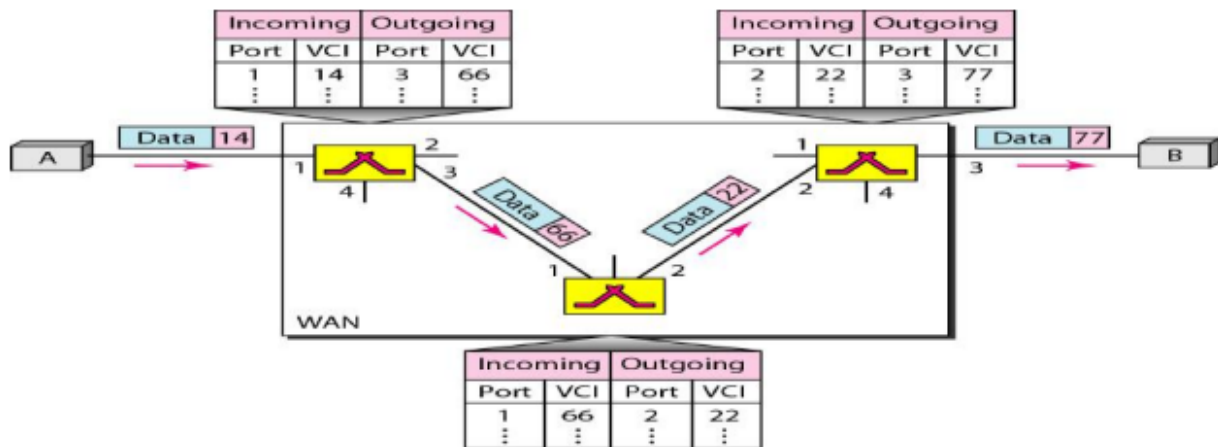


Three Phases Three phases in a virtual-circuit network: setup, data transfer, and teardown. We first discuss the data transfer phase, which is more straightforward; we then talk about the setup and teardown phases.

Data Transfer Phase:



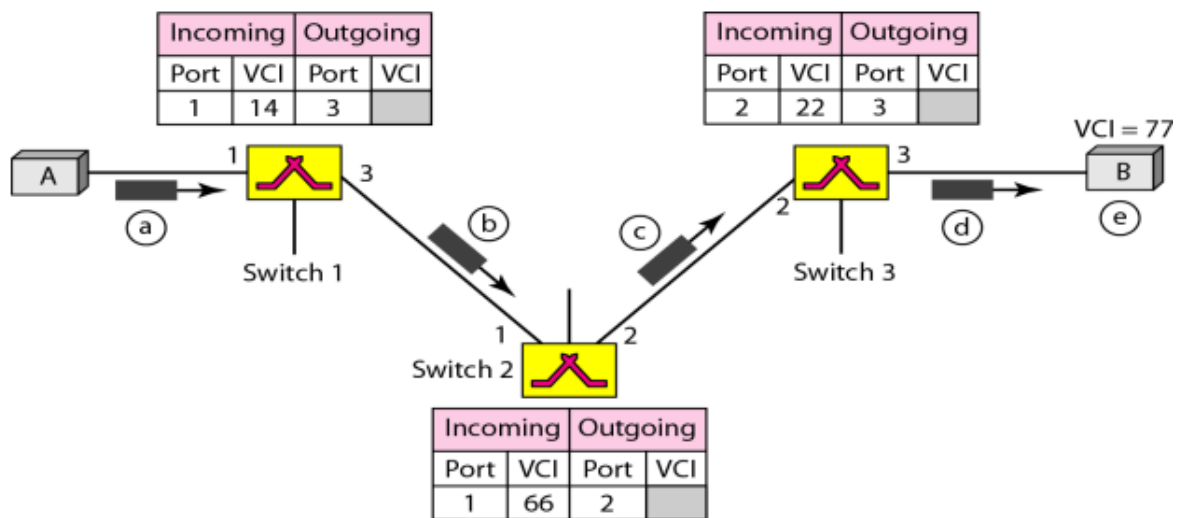
To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit. The table, in its simplest form, has four columns. We show later how the switches make their table entries, but for the moment we assume that each switch has a table with entries for all active virtual circuits. Figure shows such a switch and its corresponding table. Figure shows a frame arriving at port 1 with a VCI of 14. When the frame arrives, the switch looks in its table to find port 1 and a VCI of 14. When it is found, the switch knows to change the VCI to 22 and send out the frame from port 3. Figure shows how a frame from source A reaches destination B and how its VCI changes during the trip



Each switch changes the VCI and routes the frame. The data transfer phase is active until the source sends all its frames to the destination. The procedure at the switch is the same for each frame of a message. The process creates a virtual circuit, not a real circuit, between the source and destination.

Setup Phase:

In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the setup request and the acknowledgment. Setup Request A setup request frame is sent from the source to the destination. Figure shows the process.



a. Source A sends a setup frame to switch 1.

b. Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. For the moment, assume that it knows the output port. The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the

outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.

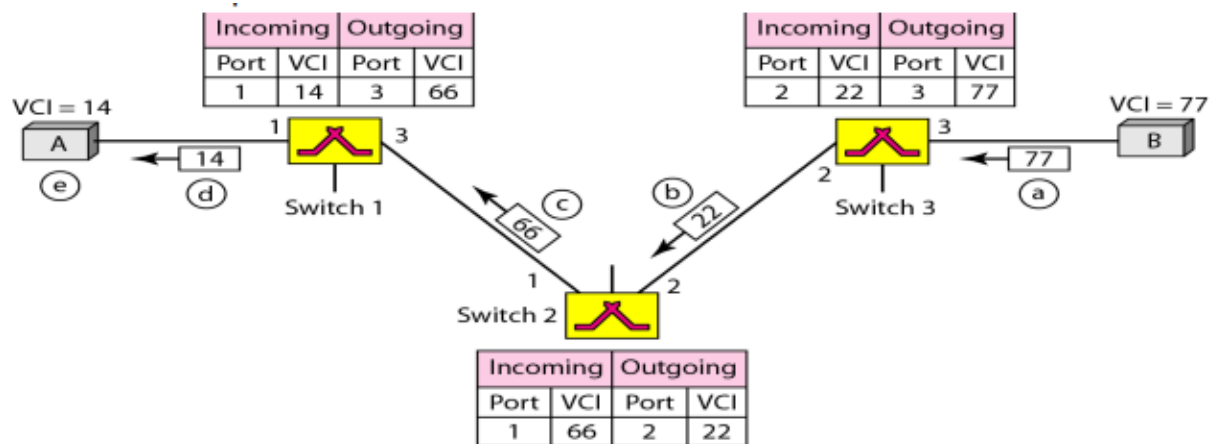
c. Switch 2 receives the setup request frame. The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).

d. Switch 3 receives the setup request frame. Again, three columns are completed: incoming port (2), incoming VCI (22), and outgoing port (3).

e. Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and not other sources.

Acknowledgment:

A special frame, called the acknowledgment frame, completes the entries in the switching tables. Figure shows the process



a. The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed. The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.

b. Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.

- c. Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.
- d. Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.
- e. The source uses this as the outgoing VCI for the data frames to be sent to destination B.

Teardown Phase :

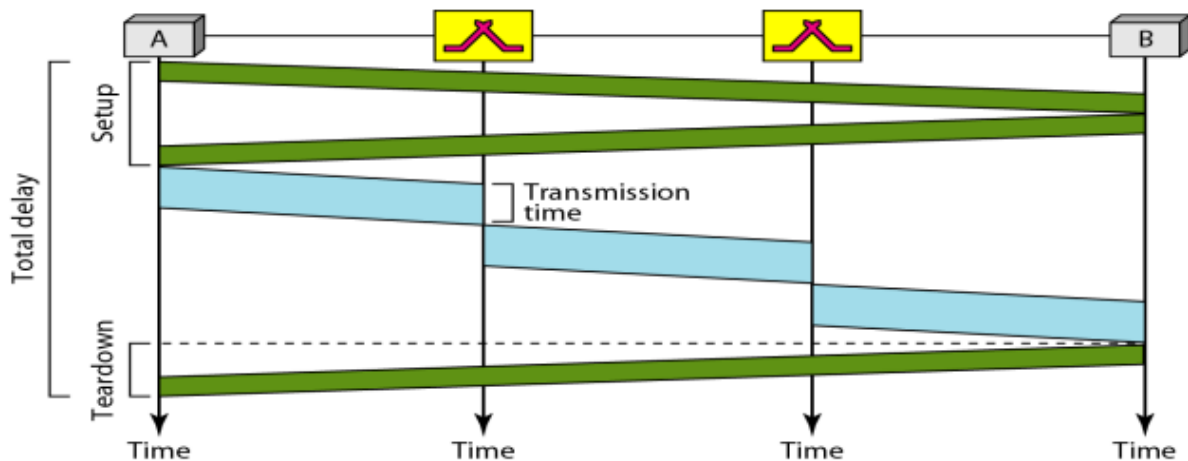
In this phase, source A, after sending all frames to B, sends a special frame called a teardown request. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables

Efficiency :

In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Delay :

In a virtual-circuit network, there is a one-time delay for setup and a one-time delay for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets. Figure shows the delay for a packet traveling through two switches in a virtual-circuit network.



Switching at the data link layer in a switched WAN is normally implemented by using virtual-circuit techniques.

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

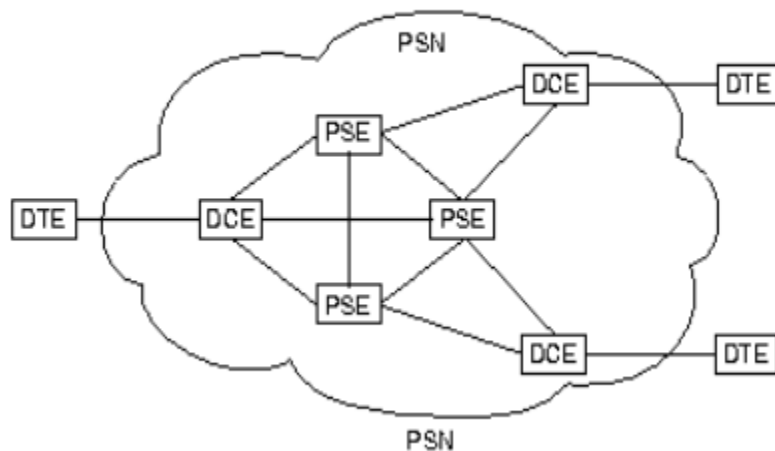
X.25:

A packet-switching protocol for wide area network (WAN) connectivity that uses a public data network (PDN) that parallels the voice network of the Public Switched Telephone Network (PSTN). The current X.25 standard supports synchronous, full-duplex communication at speeds up to 2 Mbps over two pairs of wires, but most implementations are 64-Kbps connections via a standard DS0 link.

X.25 defines a telephone network for data communications.

To begin communication, one computer calls another to request a communication session. The called computer can accept or refuse the connection. If the call is accepted, the two systems can begin full-duplex information transfer. Either side can terminate the connection at any time.

The X.25 specification defines a point-to-point interaction between data terminal equipment (DTE) and data communication equipment (DCE). DTEs (terminals and hosts in the user's facilities) connect to DCEs (modems, packet switches, and other ports into the PDN, generally located in the carrier's facilities), which connect to packet switching exchanges (PSEs, or simply switches) and other DCEs inside a PSN and, ultimately, to another DTE. The relationship between the entities in an X.25 network is shown in Figure

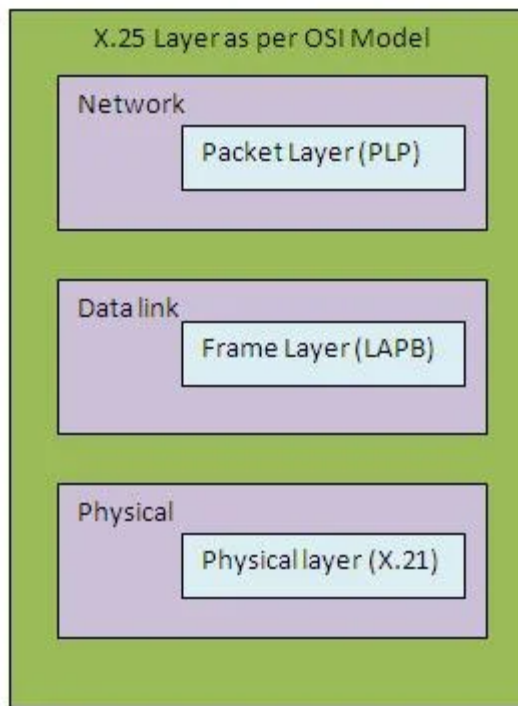


Because X.25 was designed when analog telephone transmission over copper wire was the norm, X.25 packets have a relatively large overhead of error-correction information, resulting in comparatively low overall bandwidth. Newer WAN technologies such as frame relay, Integrated Services Digital Network (ISDN), and Tcarrier services are now generally preferred over X.25. However, X.25 networks still have applications in areas such as credit card verification, automatic teller machine transactions, and other dedicated business and financial uses

How It Works:

The X.25 standard corresponds in functionality to the first three layers of the Open Systems Interconnection (OSI) reference model for networking. Specifically, X.25 defines the following:

- The physical layer interface for connecting data terminal equipment (DTE), such as computers and terminals at the customer premises, with the data communications equipment (DCE), such as X.25 packet switches at the X.25 carrier's facilities. The physical layer interface of X.25 is called X.21bis and was derived from the RS-232 interface for serial transmission.
- The data-link layer protocol called Link Access Procedure, Balanced (LAPB), which defines encapsulation (framing) and error-correction methods. LAPB also enables the DTE or the DCE to initiate or terminate a communication session or initiate data transfer. LAPB is derived from the Highlevel Data Link Control (HDLC) protocol.
- The network layer protocol called the Packet Layer Protocol (PLP), which defines how to address and deliver X.25 packets between end nodes and switches on an X.25 network using permanent virtual circuits (PVCs) or switched virtual circuits (SVCs). This layer is responsible for call setup and termination and for managing transfer of packets.



An X.25 network consists of a backbone of X.25 switches that are called packet switching exchanges (PSEs). These switches provide packet-switching services that connect DCEs at the local facilities of X.25 carriers. DTEs at customer premises connect to DCEs at X.25 carrier facilities by using a device called a packet assembler/disassembler (PAD). You can connect several DTEs to a single DCE by using the multiplexing methods inherent in the X.25 protocol. Similarly, a single X.25 end node can establish several virtual circuits simultaneously with remote nodes. An end node (DTE) can initiate a communication session with another end node by dialing its X.121 address and establishing a virtual circuit that can be either permanent or switched, depending on the level of service required. Packets are routed through the X.25 backbone network by using the ID number of the virtual circuit established for the particular communication session. This ID number is called the logical channel identifier (LCI) and is a 12-bit address that identifies the virtual circuit. Packets are generally up to 128 bytes in size, although maximum packet sizes range from 64 to 4096 bytes, depending on the system.

Disadvantages of X.25:

Prior to Frame Relay, some organizations were using a virtual-circuit switching network called X.25 that performed switching at the network layer. For example, the Internet, which needs wide-area networks to carry its packets from one place to another, used X.25. And X.25 is still being used by the Internet, but it is being replaced by other WANs.

However, X.25 has several drawbacks:

- X.25 has a low 64-kbps data rate. By the 1990s, there was a need for higher- data-rate WANs.

- X.25 has extensive flow and error control at both the data link layer and the network layer. This was so because X.25 was designed in the 1970s, when the available transmission media were more prone to errors. Flow and error control at both layers create a large overhead and slow down transmissions. X.25 requires acknowledgments for both data link layer frames and network layer packets that are sent between nodes and between source and destination.

- Originally X.25 was designed for private use, not for the Internet. X.25 has its own network layer. This means that the user's data are encapsulated in the network layer packets of X.25. The Internet, however, has its own network layer, which means if the Internet wants to use X.25, the Internet must deliver its network layer packet, called a datagram, to X.25 for encapsulation in the X.25 packet. This doubles the overhead.

Frame Relay:

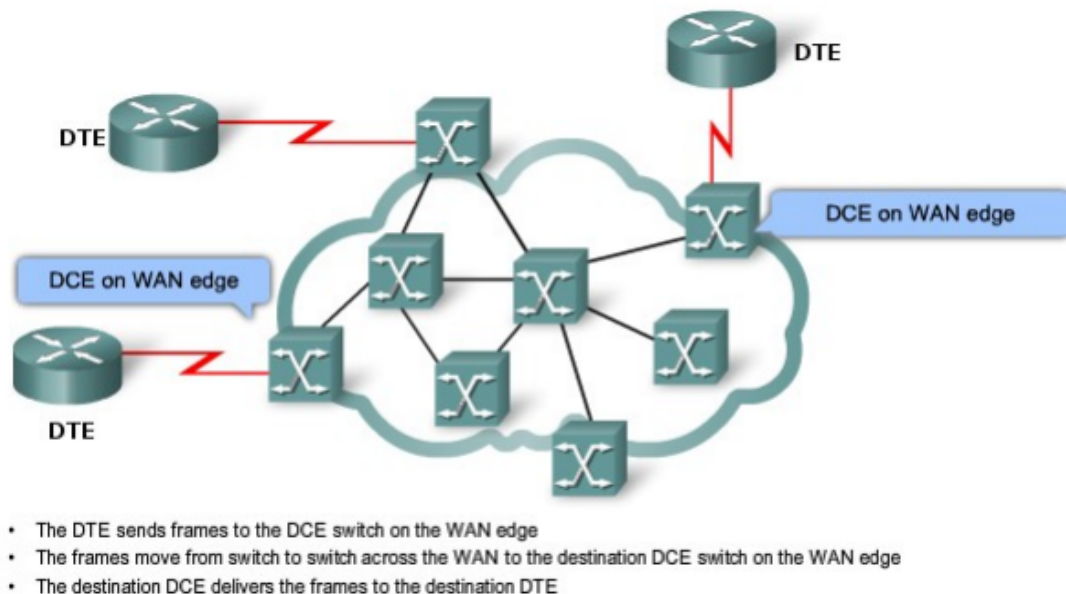
Frame Relay is a high-performance WAN protocol that operates at the physical and Data Link layers of the OSI reference model. X.25 has several disadvantages so Frame Relay was invented.

Frame Relay is a wide-area network with the following features:

1. Frame Relay operates at a higher speed (1.544 Mbps and recently 44.376 Mbps). This means that it can easily be used instead of a mesh of T-1 or T-3 lines.
2. Frame Relay operates in just the physical and data link layers. This means it can easily be used as a backbone network to provide services to protocols that already have a network layer protocol, such as the Internet.
3. Frame Relay allows bursty data.
4. Frame Relay allows a frame size of 9000 bytes, which can accommodate all local- area network frame sizes.
5. Frame Relay is less expensive than other traditional WANs.
6. Frame Relay has error detection at the data link layer only. There is no flow control or error control. There is not even a retransmission policy if a frame is damaged; it is silently dropped. Frame Relay was designed in this way to provide fast transmission capability for more reliable media and for those protocols that have flow and error control at the higher layers.

Frame Relay Operation:

When carriers use Frame Relay to interconnect LANs, a router on each LAN is the DTE. A serial connection, such as a T1/E1 leased line, connects the router to the Frame Relay switch of the carrier at the nearest point-of-presence (POP) for the carrier. The Frame Relay switch is a DCE device. Network switches move frames from one DTE across the network and deliver frames to other DTEs by way of DCEs.



Virtual Circuits : The connection through a Frame Relay network between two DTEs is called a virtual circuit (VC). The circuits are virtual because there is no direct electrical connection from end to end. The connection is logical, and data moves from end to end, without a direct electrical circuit. With VCs, Frame Relay shares the bandwidth among multiple users and any single site can communicate with any other single site without using multiple dedicated physical lines.

There are two ways to establish VCs:

Permanent Virtual Circuit(PVC):

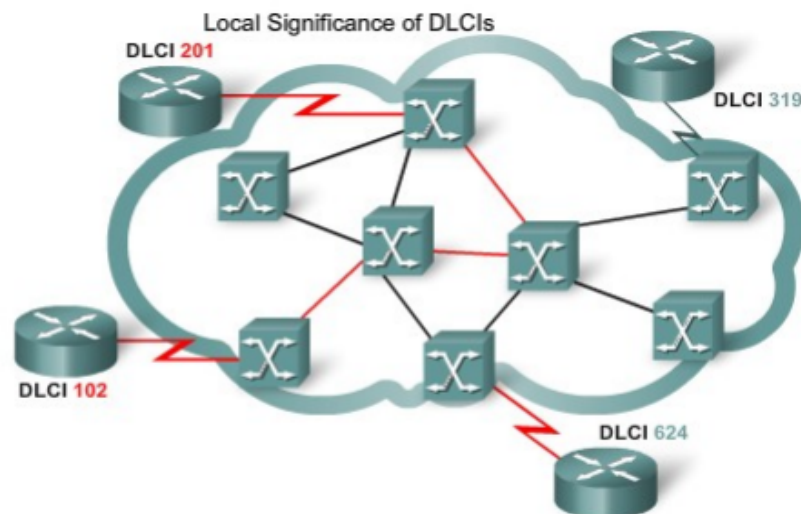
A source and a destination may choose to have a permanent virtual circuit (PVC). In this case, the connection setup is simple. The corresponding table entry is recorded for all switches by the administrator (remotely and electronically, of course). An outgoing DLCI is given to the source, and an incoming DLCI is given to the destination. PVC connections have two drawbacks. First, they are costly because two parties pay for the connection all the time even when it is not in use. Second, a connection is created from one source to one single destination. If a source needs connections with several destinations, it needs a PVC for each connection. PVCs, permanent virtual circuits, are preconfigured by the carrier, and after they are set up, only operate in DATA TRANSFER and IDLE modes. Note that some publications refer to PVCs as private Vcs.

Switched Virtual-circuit(SVC):

An alternate approach is the switched virtual circuit (SVC). The SVC creates a temporary, short connection that exists only when data are being transferred between source and destination. SVCs, switched virtual circuits, are established dynamically by sending signaling messages to the network (CALL SETUP, DATA TRANSFER, IDLE, CALL TERMINATION).

DLCI (Data link connection identifier):

Frame Relay is a virtual circuit network. A virtual circuit in Frame Relay is identified by a number called a data link connection identifier (DLCI). DLCI values typically are assigned by the Frame Relay service provider (for example, the telephone company). Usually, DLCIs 0 to 15 and 1008 to 1023 are reserved for special purposes. Therefore, service providers typically assign DLCIs in the range of 16 to 1007. Frame Relay DLCIs have local significance, which means that the values themselves are not unique in the Frame Relay WAN. A DLCI identifies a VC to the equipment at an endpoint. A DLCI has no significance beyond the single link. Two devices connected by a VC may use a different DLCI value to refer to the same connection.



DLCI values have local significance, which means that they are unique only to the physical channel on which they reside. Therefore, devices at opposite ends of a connection can use the same DLCI values to refer to different virtual circuits.

Frame Relay Layers:

Frame Relay operates at the physical layer and the Data link layer.

Physical Layer No specific protocol is defined for the physical layer in Frame Relay. Instead, it is left to the implementer to use whatever is available. Frame Relay supports any of the protocols recognized by ANSI. **Data Link Layer** At the data link layer, Frame Relay uses a simple protocol that does not support flow or error control. It only has an error detection mechanism. Figure below shows the format of a Frame Relay frame. The address field defines the DLCI as well as some bits used to control congestion.

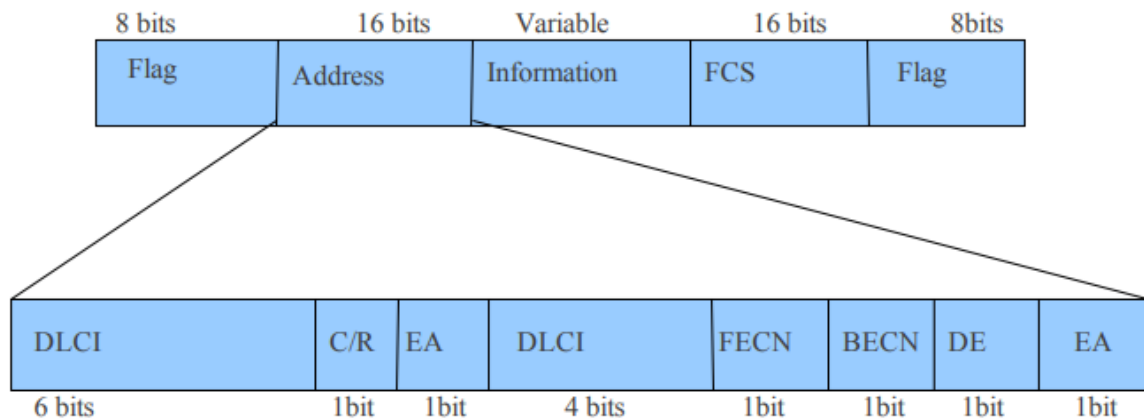


Fig:Frame Relay Frame Format

C/R: Command/response

EA: Extended address

FECN: Forward explicit congestion notification

BECN: Backward explicit congestion notification

DE: Discard eligibility

DLCI: Data link connection identifier

Address (DLCI) field.

The first 6 bits of the first byte makes up the first part of the DLCI. The second part of the DLCI uses the first 4 bits of the second byte. These bits are part of the 10-bit data link connection identifier defined by the standard.

Command/response (C/R).

The command/response (C/R) bit is provided to allow upper layers to identify a frame as either a command or a response. It is not used by the Frame Relay protocol.

Extended address (EA).

The extended address (EA) bit indicates whether the current byte is the final byte of the address. An EA of 0 means that another address byte is to follow. An EA of 1 means that the current byte is the final one.

Forward explicit congestion notification (FECN).

The forward explicit congestion notification (FECN) bit can be set by any switch to indicate that traffic is congested. This bit informs the destination that congestion has occurred. In this way, the destination knows that it should expect delay or a loss of packets.

Backward explicit congestion notification (BECN).

The backward explicit congestion notification (BECN) bit is set (in frames that travel in the other direction) to indicate a congestion problem in the network. This bit informs the sender that congestion has occurred. In this way, the source knows it needs to slow down to prevent the loss of packets .

Discard eligibility (DE).

The discard eligibility (DE) bit indicates the priority level of the frame. In emergency situations, switches may have to discard frames to relieve bottlenecks and keep the network from collapsing due to overload. When set (DE 1), this bit tells the network to discard this frame if there is congestion. This bit can be set either by the sender of the frames (user) or by any switch in the network.

Asynchronous Transfer Mode (ATM) :

Asynchronous transfer mode (ATM), also known as cell relay, is similar in concept to frame relay. Both frame relay and ATM take advantage of the reliability and fidelity of modern digital facilities to provide faster packet switching than X.25

ATM is even more streamlined than frame relay in its functionality, and can support data rates several orders of magnitude greater than frame relay

The “asynchronous” in ATM means ATM devices do not send and receive information at fixed speeds or using a timer, but instead negotiate transmission speeds based on hardware and information flow reliability. The “transfer mode” in ATM refers to the fixed-size cell structure used for packaging information. ATM transfers information in fixed-size units called cells. Each cell consists of 53 octets, or bytes as shown in Fig



Fig:ATM cell Format

- Transmits all information in fixed size blocks called cells.
- Cells are transmitted asynchronously.
- The network is connection oriented.
- Each cell is 53 bytes long – 5 bytes header and 48 bytes payload.
- Making an ATM call requires first sending a message to set up a connection. Subsequently all cells follow the same path to the destination
- ATM was envisioned as the technology for providing B-ISDN services.
- It can handle both constant rate traffic and variable-length traffic. Thus, it can carry multiple types of traffic with end-to-end quality of service.
- ATM is independent of transmission medium. It doesn't prescribe any particular rule.

- They may be sent on a wire or Fiber by themselves or they may be also packaged inside the payload of the other carrier system.
- Delivery of the system is not guaranteed but the order is.

When the virtual circuit is established, what really happens is that a route is chosen from source to destination. All the switches along the way make table entries for the virtual circuit and have the opportunity to reserve resources for the new circuit. The cells are sent from one switch to the next (stored and forwarded) until they reach the destination. When a cell comes along, the switch inspects its header to find out which virtual circuit it belongs to.

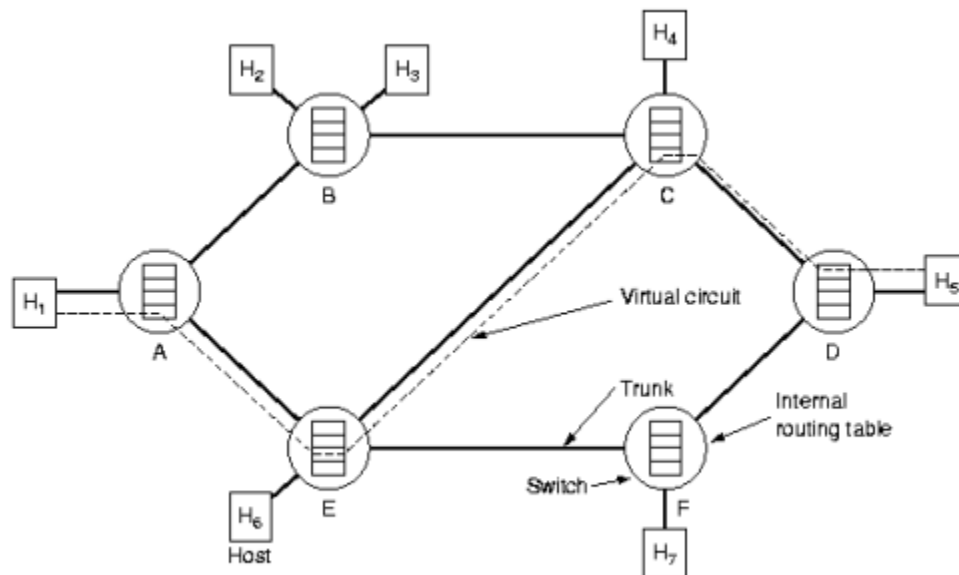


Fig: Cells Traveling from Host#1 to Host#5:-

ATM Network Interfaces : An ATM network consists of a set of ATM switches interconnected by point-to-point ATM links or interfaces. ATM switches support two primary types of interfaces: UNI and NNI as shown in Fig. Below.

The **UNI (User Network Interface)** connects ATM end systems (such as hosts and routers) to an ATM switch.

The **NNI (Network-Network Interface)** connects two ATM switches. Depending on whether the switch is owned and located at the customer's premises or is publicly owned and operated by the telephone company, UNI and NNI can be further subdivided into public and private UNIs and NNIs. A private UNI connects an ATM endpoint and a private ATM switch. Its public counterpart connects an ATM endpoint or private switch to a public switch. A private NNI

connects two ATM switches within the same private organization. A public one connects two ATM switches within the same public organization

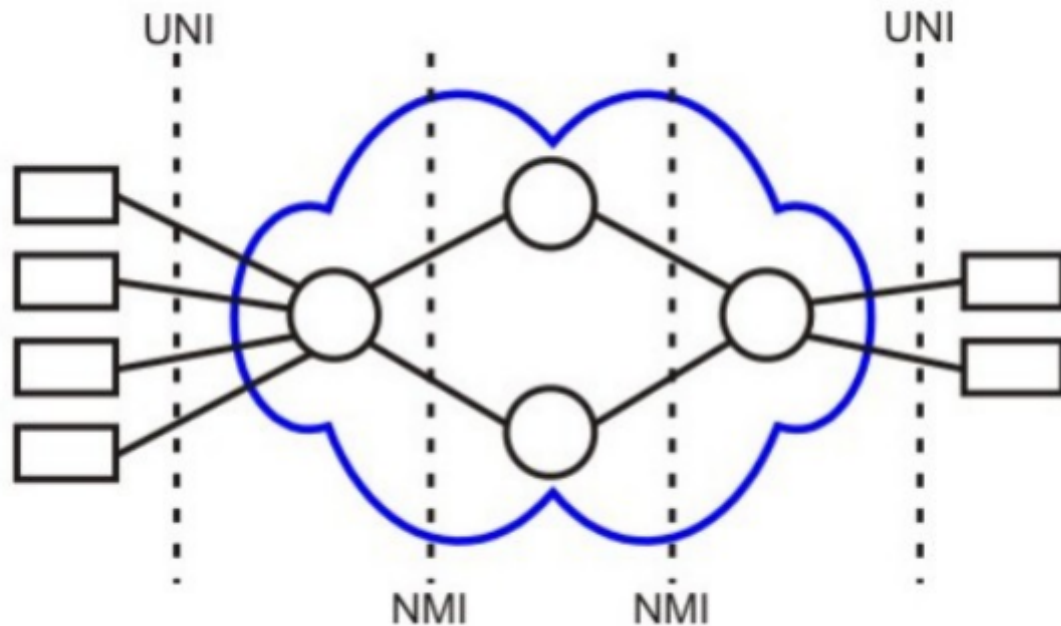


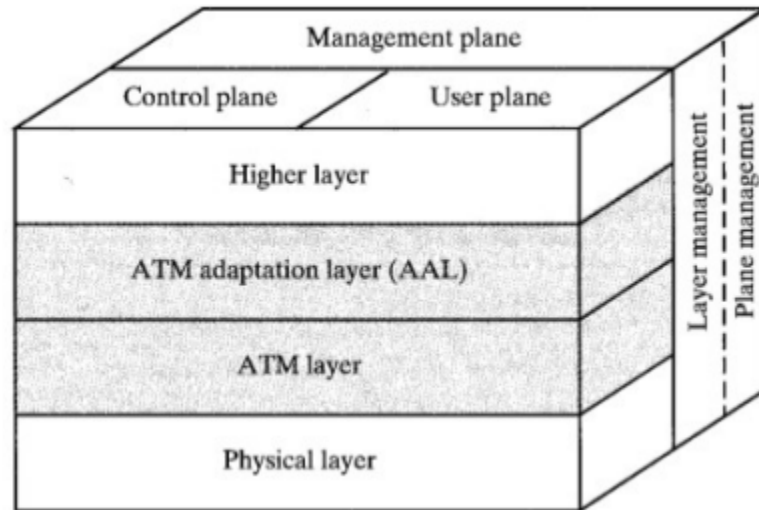
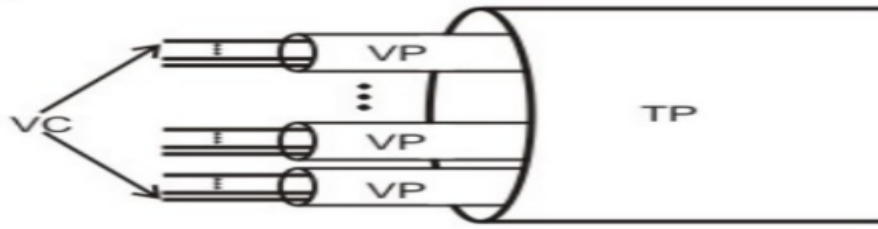
Fig: UNI and NNI interfaces of the ATM

ATM Virtual Connections ATM operates as a channel-based transport layer, using Virtual circuits (VCs). This is encompassed in the concept of the Virtual Paths (VP) and Virtual Channels. Every ATM cell has an 8- or 12-bit Virtual Path Identifier (VPI) and 16-bit Virtual Channel Identifier (VCI) pair defined in its header. Together, these identify the virtual circuit used by the connection. The length of the VPI varies according to whether the cell is sent on the user-network interface (on the edge of the network), or if it is sent on the network-network interface (inside the network).

As these cells traverse an ATM network, switching takes place by changing the VPI/VCI values (label swapping). Although the VPI/VCI values are not necessarily consistent from one end of the connection to the other, the concept of a circuit is consistent (unlike IP, where any given packet could get to its destination by a different route than the others).

Another advantage of the use of virtual circuits comes with the ability to use them as a multiplexing layer, allowing different services (such as voice, Frame Relay, n* 64 channels, IP).

A virtual path connection (VPC) is a bundle of VCCs that have the same endpoints. Thus, all of the cells flowing over all of the VCCs in a single VPC are switched together



VC- Virtual Circuit VP- Virtual Path TP-

Transmission Path ATM Reference Model: The protocol reference model makes reference to three separate planes:

User Plane.

Provides for user information transfer, along with associated controls (e.g., flow control, error control).

Control Plane.

Performs call control and connection control functions.

Management Plane.

Includes plane management, which performs management functions related to a system as a whole and provides coordination between all the planes, and layer management, which performs management functions relating to resources and parameters residing in its protocol entities.

Physical layer—Analogous to the physical layer of the OSI reference model, the ATM physical layer manages the medium-dependent transmission.

ATM layer—Combined with the ATM adaptation layer, the ATM layer is roughly analogous to the data link layer of the OSI reference model. The ATM layer is responsible for the simultaneous sharing of virtual circuits over a physical link (cell multiplexing) and passing cells through the ATM network (cell relay). To do this, it uses the VPI and VCI information in the header of each ATM cell.

ATM adaptation layer (AAL)—Combined with the ATM layer, the AAL is roughly analogous to the data link layer of the OSI model. The AAL is responsible for isolating higher-layer protocols from the details of the ATM processes. The adaptation layer prepares user data for conversion into cells and segments the data into 48-byte cell payloads

ATM Advantages:

- ATM supports voice, video and data allowing multimedia and mixed services over a single network.
- high evolution potential, works with existing, legacy technologies
- provides the best multiple service support
- supports delay close to that of dedicated services
- supports the broadest range of burstiness, delay tolerance and loss performance through the implementation of multiple QoS classes
- provides the capability to support both connection-oriented and connectionless traffic using AALs
- able to use all common physical transmission paths (such as DS1, SONET).
- cable can be twisted-pair, coaxial or fiber-optic
- ability to connect LAN to WAN
- legacy LAN emulation
- efficient bandwidth use by statistical multiplexing
- scalability
- higher aggregate bandwidth
- high speed Mbps and possibly Gbps

ATM disadvantages

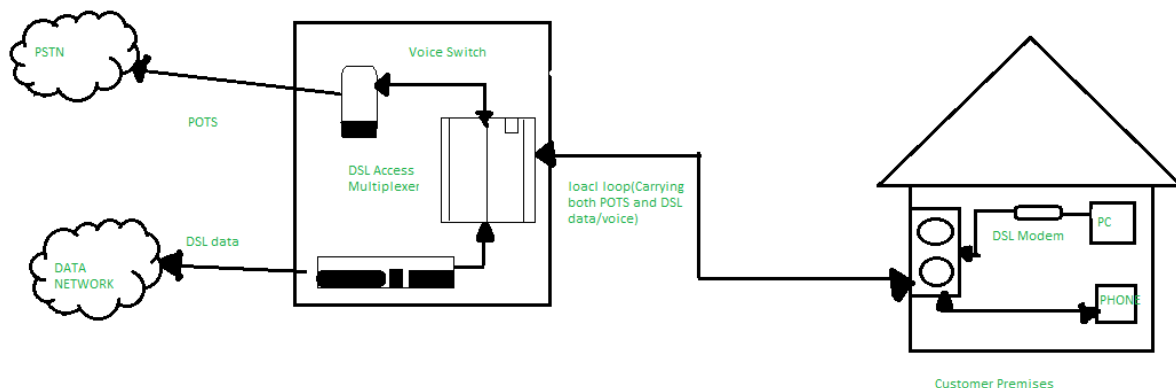
- flexible to efficiency's expense, at present, for any one application it is usually possible to find a more optimized technology
- cost, although it will decrease with time
- new customer premises hardware and software are required
- competition from other technologies -100 Mbps FDDI, 100 Mbps Ethernet and fast Ethernet.

Digital Subscriber Line (DSL)

The Digital Subscriber Line (DSL), *originally*, a **digital subscriber loop** is a communication medium, which is used to transfer the internet through copper wire telecommunication lines. Along with cable internet, DSL is one of the most popular ways *ISPs* provide broadband internet access.

Properties of DSL

- Its aim is to maintain the high speed of the data being transferred.
- If we ask how we going to achieve such a thing i.e., both telephone and internet facilities, then the answer is by using *splitters* or *DSL filters*(shown in the below diagram). Basically, the *splitter* is used to split the frequency and make sure that it can't get interrupted.



Types of DSL

Symmetric DSL

SDSL, *splits* the upstream and downstream frequencies evenly, providing equal speeds to both uploading and downloading data transfer. This connection may provide *2 Mbps* upstream and downstream. It is mostly preferred by small organizations.

Asymmetric DSL

ADSL, provides a wider frequency range for downstream transfers, which offers several times faster downstream speeds. an ADSL connection may offer *20 Mbps downstream and 1.5 Mbps upstream*, it is because most users download more data than they upload.

Features of DSL

- **Speed:** DSL provides high-speed internet connectivity that is much faster than traditional dial-up connections. With DSL, users can expect download speeds of up to 25 Mbps and upload speeds of up to 3 Mbps, depending on the type of DSL service and the distance from the service provider.
- **Availability:** DSL is widely available in many areas and is often the most accessible high-speed internet option for users who live in rural or remote areas. As long as there is a telephone line, DSL can be provided, making it an attractive option for users who don't have access to cable or fiber internet.
- **Security:** DSL provides a secure connection, as the data transmitted over the telephone lines is encrypted. This means that hackers and other unauthorized users cannot intercept or access the data being transmitted.
- **Reliability:** DSL is a reliable internet connection option, as the connection is always on and doesn't require users to dial in or connect manually. This makes it ideal for users who need a constant internet connection for work or other purposes.
- **Affordability:** DSL is generally more affordable than other high-speed internet options, such as cable or fiber internet. This makes it an attractive option for users who need high-speed internet but don't want to pay the high prices associated with other options.
- **Compatibility:** DSL is compatible with a wide range of devices, including computers, smartphones, tablets, and other internet-connected devices. This makes it easy for users to connect to the internet and access online services and applications.

Benefits

- **No Additional Wiring** – A DSL connection makes use of your existing telephone wiring, so you will not have to pay for expensive upgrades to your phone system.
- **Cost-Effective** – DSL internet is a very cost-effective method and is best in connectivity
- Availability of DSL modems by the service providers.
- Users can use both telephone lines and the internet at the same time. And it is because the voice and digital signals are transferred in different frequencies.
- Users can choose between different connection *speeds* and *pricing* from various providers.
- **High-speed:** DSL provides high-speed internet access, allowing users to stream, download, and upload large amounts of data quickly.
- **Simultaneous use:** DSL allows users to make voice calls and use the internet at the same time, making it a convenient option for households with multiple users who need to use the internet and the telephone at the same time.
- **Reliable:** DSL is a stable and reliable technology that provides consistent performance, making it a dependable option for internet access.

- **No data caps:** Unlike some other internet technologies, DSL doesn't have data caps, allowing users to stream, download, and upload as much data as they need without worrying about additional charges or throttled speeds.
- **Easy setup:** DSL is easy to set up and doesn't require any special equipment, making it a straightforward option for people who are new to the internet or need to get online quickly.
- **Supports multiple devices:** DSL can support multiple devices simultaneously, making it an ideal option for households with multiple users and devices.

DSL Internet service only works over a limited physical distance and remains unavailable in many areas where the local telephone infrastructure does not support DSL technology. The service is not available everywhere. The connection is faster for receiving data than it is for sending data over the Internet

What is a Cable Modem?

A cable modem can be used to access the Internet on a computer at home or at work while using a cable TV connection. They frequently function more swiftly than DSL modems, which connect computers to the Internet using phone lines. A cable modem can be used with any Internet service, including dial-up, DSL, and cable TV.

A cable modem is a device used to connect a computer or router to a cable television line which provides the high-speed data connection for cable Internet service. A cable modem typically uses an Ethernet cable to connect to the computer or router.

Characteristics	DSL Modem	Cable Modem
Definition	DSL is a type of broadband Internet connection that delivers fast speeds over a standard copper phone line. A DSL modem connects to your phone line and then to your router or computer.	A cable modem connects to your TV's coaxial input and then to your router or computer.
Type of Connection	Uses the telephone lines	Uses the Cable TV Coax
Speeds	Range up to 300 Mbps	Range up to 10 Gbps
Connectivity	DSL modem contains the dedicated signal	Cable modem contains the shared signal

Stability	It is more stable due to the landline	It is less stable as compared to Cable Modem
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So, Which is Better? DSL Modem or Cable Modem

DSL uses a phone line to deliver internet service, while cable uses a coaxial cable, or “coax,” similar to the type used for cable TV service. Now that we know how each type of internet service is delivered, let’s explore the key differences between DSL and cable –

- **Speed** – Cable offers faster speeds than DSL. If you download a lot of content or stream video regularly, you’ll want to consider a plan with higher speeds.
- **Availability** – DSL is widely available across the U.S., but availability varies by location. Cable is also widely available, but in some areas only one ISP may offer service.
- **Price** – In general, DSL plans are cheaper than comparable cable plans. However, pricing can vary depending on your location and the speed of the plan you select.

The distinction between a cable modem and a DSL modem is pretty obvious. Since cable modems offer faster download rates than DSL modems, they are the most popular type of modem. However, a DSL modem can be a better option for you if you require a more dependable connection without bandwidth limits.

Both DSL and cable modems use different frequency bands to transmit data. **DSL modulation uses lower frequencies**, which means that it can be transmitted over longer distances with less

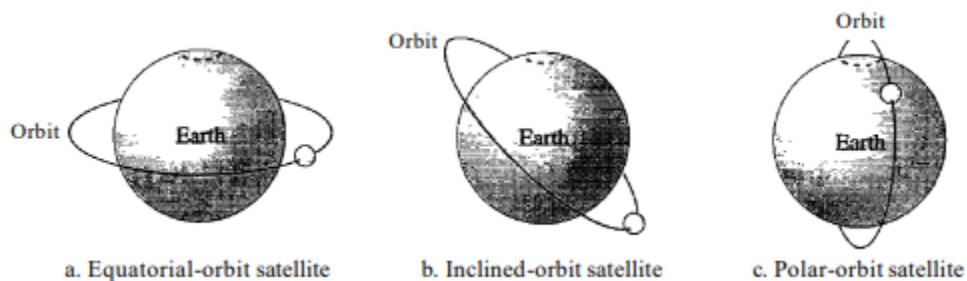
signal degradation. This also means that more people can subscribe to a single DSL connection without affecting speeds. **Cable modulation, on the other hand, uses higher frequencies,** which limits the distance it can travel but provides faster speeds.

SATELLITE NETWORKS:

A satellite network is a combination of nodes, some of which are satellites, that provides communication from one point on the Earth to another. A node in the network can be a satellite, an Earth station, or an end-user terminal or telephone. Although a natural satellite, such as the Moon, can be used as a relaying node in the network, the use of artificial satellites is preferred because we can install electronic equipment on the satellite to regenerate the signal that has lost its energy during travel. Another restriction on using natural satellites is their distances from the Earth, which create a long delay in communication. Satellite networks are like cellular networks in that they divide the planet into cells. Satellites can provide transmission capability to and from any location on Earth, no matter how remote. This advantage makes high-quality communication available to undeveloped parts of the world without requiring a huge investment in ground-based infrastructure

Orbits :

An artificial satellite needs to have an orbit~ the path in which it travels around the Earth. The orbit can be equatorial, inclined, or polar, as shown in Figure

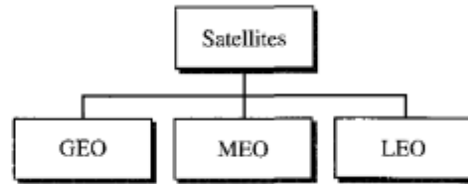


The period of a satellite, the time required for a satellite to make a complete trip around the Earth, is determined by Kepler's law, which defines the period as a function of the distance of the satellite from the center of the Earth

Three Categories of Satellites Based on the location of the orbit, satellites can be divided into three categories:

geostationary Earth orbit (GEO),

low-Earth-orbit (LEO),
and middle-Earth-orbit (MEO).



GEO Satellites :

Line-of-sight propagation requires that the sending and receiving antennas be locked onto each other's location at all times (one antenna must have the other in sight). For this reason, a satellite that moves faster or slower than the Earth's rotation is useful only for short periods. To ensure constant communication, the satellite must move at the same speed as the Earth so that it seems to remain fixed above a certain spot. Such satellites are called geostationary.

Because orbital speed is based on the distance from the planet, only one orbit can be geostationary. This orbit occurs at the equatorial plane and is approximately 22,000 mi from the surface of the Earth. But one geostationary satellite cannot cover the whole Earth.

One satellite in orbit has line-of-sight contact with a vast number of stations, but the curvature of the Earth still keeps much of the planet out of sight. It takes a minimum of three satellites equidistant from each other in geostationary Earth orbit (GEO) to provide full global transmission

MEO Satellites :

Medium-Earth-orbit (MEO) satellites are positioned between the two Van Allen belts. A satellite at this orbit takes approximately 6-8 hours to circle the Earth.

Global Positioning System One example of a MEO satellite system is the Global Positioning System (GPS), constructed and operated by the US Department of Defense, orbiting at an altitude about

18,000 km (11,000 mi) above the Earth. The system consists of 24 satellites and is used for land, sea, and air navigation to provide time and locations for vehicles and ships.

The orbits and the locations of the satellites in each orbit are designed in such a way that, at any time, four satellites are visible from any point on Earth. A GPS receiver has an almanac that tells the current position of each satellite

Application :

GPS is used by military forces. For example, thousands of portable GPS receivers were used during the Persian Gulf war by foot soldiers, vehicles, and helicopters. Another use of GPS is in navigation. The driver of a car can find the location of the car. The driver can then consult a database in the memory of the automobile to be directed to the destination. In other words, GPS gives the location of the car, and the database uses this information to find a path to the destination. A very interesting application is clock synchronization. As we mentioned previously, the IS-95 cellular telephone system uses GPS to create time synchronization between the base stations.

LEO Satellites:

Low-Earth-orbit (LEO) satellites have polar orbits. The altitude is between 500 and 2000 km, with a rotation period of 90 to 120 min. The satellite has a speed of 20,000 to 25,000 km/h. An LEO system usually has a cellular type of access, similar to the cellular telephone system. The footprint normally has a diameter of 8000 km. Because LEO satellites are close to Earth, the round-trip time propagation delay is normally less than 20 ms, which is acceptable for audio communication. An LEO system is made of a constellation of satellites that work together as a network; each satellite acts as a switch. Satellites that are close to each other are connected through inter satellite links (ISLs). A mobile system communicates with the satellite through a user mobile link (UML). A satellite can also communicate with an Earth station (gateway) through a gateway link (GWL). Figure 16.19 shows a typical LEO satellite network.

LEO satellites can be divided into three categories:

little LEOs, big LEOs, and broadband LEOs.

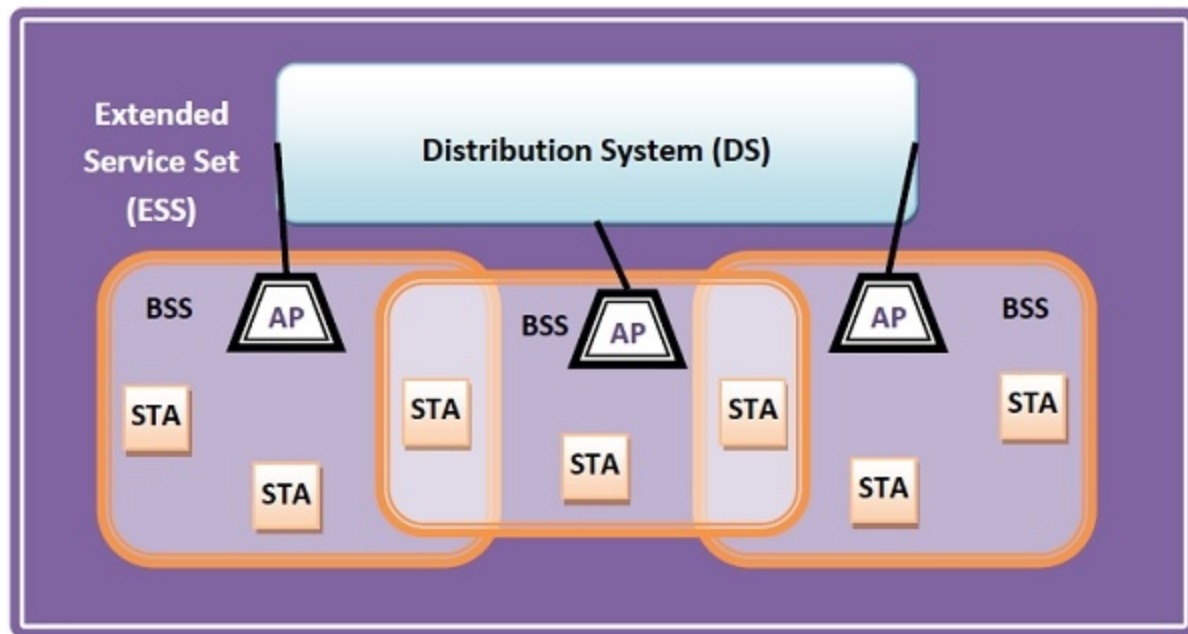
The little LEOs operate under 1 GHz. They are mostly used for low-data-rate messaging.

The big LEOs operate between 1 and 3 GHz. Globalstar and Iridium systems are examples of big LEOs. The broadband LEOs provide communication similar to fiberoptic networks. The first broadband LEO system was Teledesic.

IEEE 802.11 Architecture

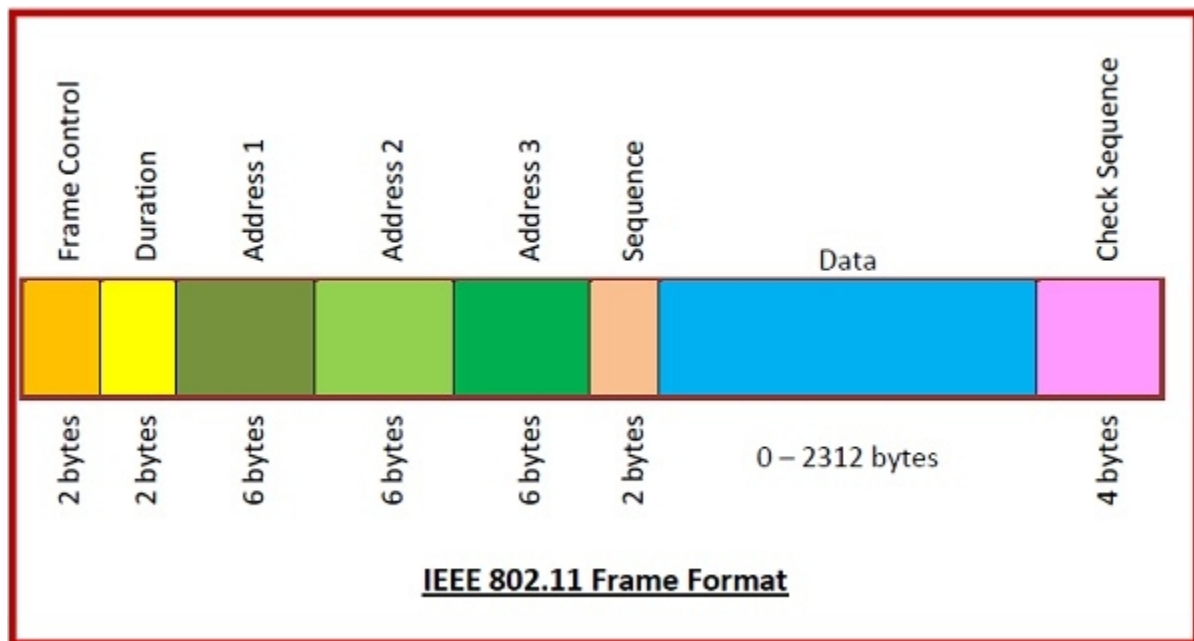
IEEE 802.11 standard, popularly known as **WiFi**, lays down the architecture and specifications of **wireless LANs (WLANs)**. WiFi or WLAN uses high-frequency radio waves instead of cables for connecting the devices in **LAN**. Users connected by WLANs can move around within the area of network coverage

Important Terminologies of IEEE 802.11 Architecture



- **Stations (STA)** – Stations comprises of all devices and equipment that are connected to the wireless LAN. A station can be of two types–
 - Wireless Access Point (WAP) – WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.
 - Client. Clients are workstations, computers, laptops, printers, smartphones, etc.
- Each station has a wireless network interface controller.
- **Basic Service Set (BSS)** – A basic service set is a group of stations communicating at the physical layer level. BSS can be of two categories depending upon the mode of operation–
 - Infrastructure BSS – Here, the devices communicate with other devices through access points.
 - Independent BSS – Here, the devices communicate in a peer-to-peer basis in an ad hoc manner.
- **Extended Service Set (ESS)** – It is a set of all connected BSS.
- **Distribution System (DS)** – It connects access points in ESS.

Frame Format of IEEE 802.11



- **Frame Control** – It is a 2 bytes starting field composed of 11 subfields. It contains control information of the frame.
- **Duration** – It is a 2-byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.
- **Address fields** – There are three 6-byte address fields containing addresses of source, immediate destination, and final endpoint respectively.
- **Sequence** – It is a 2 bytes field that stores the frame numbers.
- **Data** – This is a variable-sized field that carries the data from the upper layers. The maximum size of the data field is 2312 bytes.
- **Check Sequence** – It is a 4-byte field containing error detection information.

IEEE 802.15:

IEEE 802.15 is a [working group](#) of the [Institute of Electrical and Electronics Engineers](#) (IEEE) [IEEE 802](#) standards committee which specifies Wireless Specialty Networks (WSN) standards. The working group was formerly known as Working Group for Wireless Personal Area Networks:

IEEE 802.15.1: WPAN / Bluetooth

Task group one is based on [Bluetooth](#) technology. It defines physical layer (PHY) and [medium access control](#) (MAC) specification for wireless connectivity with fixed, portable and moving devices within or entering personal operating space

IEEE 802.15.2: Coexistence

Task group two addresses the coexistence of wireless personal area networks ([WPAN](#)) with other wireless devices operating in unlicensed frequency bands such as wireless local area networks ([WLAN](#)).

IEEE 802.15.3: High Rate WPAN

IEEE 802.15.3-2003 is a MAC and PHY standard for high-rate (11 to 55 Mbit/s) WPANs

Applications of IEEE 802:

IEEE 802.15.4 technology finds numerous applications in various fields, including industrial automation, smart energy, healthcare, home automation, and environmental monitoring and control.

- **Industrial Automation** – This section highlights how IEEE 802.15.4 technology is an integral part of industrial automation and how wireless sensor networks utilizing IEEE 802.15.4 Technology can support a range of applications such as asset tracking, temperature control, and predictive maintenance.
- **Smart Energy** – This section discusses how IEEE 802.15.4 technology enables remote control, monitoring, and management of energy consumption by using smart meters and sensors, resulting in better demand management, lower costs, and prevention of blackouts.
- **Healthcare** – This section discusses how IEEE 802.15.4 technology enables remote patient monitoring systems, facilitates telemedicine services for remote diagnosis and treatment of various illnesses, and enables seamless communication between different medical devices like insulin pumps, pacemakers, and cardiac monitors.

- **Home Automation** – This section explains how IEEE 802.15.4 technology impacts home automation, allowing homeowners to have remote control over different devices in their homes, making life more comfortable, convenient, and secure.
- **Environmental Monitoring and Control** – This section highlights how IEEE 802.15.4 technology enables wireless sensor networks to monitor various environmental parameters like temperature, humidity, air quality, water quality, etc., and provide real-time data for analysis and effective control of the environment.

Benefits of IEEE 802

IEEE 802.15.4 technology offers benefits such as low power consumption and extended battery life, low cost and simple implementation, limited data transmission rate, and limited range and interference.

- Low Power Consumption and Extended Battery Life
- Low Cost and Simple Implementation
- Limited Data Transmission Rate
- Limited Range and Interference